



CONTACT

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TECHNICAL SKILLS

Computer Vision

OpenCV, YOLOv8, Object Detection,
Image Classification, Segmentation,
Tracking, MediaPipe, Camera Calibra-
tion, OCR, Real-time Video Processing

AI / Machine Learning

Python, NumPy, Pandas, Scikit-learn,
PyTorch, TensorFlow/Keras, ML, DL,
CNN, LSTM, Transformer, BERT, Light-
GBM, XGBoost, CatBoost

Robotics & UAV

ROS / ROS 2, ArduPilot, MAVLink, RViz,
Arduino, Servo Control, PID, Sensor Fu-
sion, IMU, LiDAR, Robot Motion Control

Tools & Systems

Git/GitHub, Docker, Linux/Ubuntu, SQL,
Jupyter Notebook, VS Code, MLOps ba-
sics

CERTIFICATIONS

Coursera

Machine Learning Specialization
Deep Learning Specialization

LANGUAGES

Vietnamese: Native
English: Basic communication, techni-
cal reading

CAREER INTERESTS

UAV Perception
Real-time Computer Vision
Autonomous Robotics
Edge AI Deployment
AI-powered Automation

NGUYEN LAM TRUONG

AI / COMPUTER VISION ENGINEER INTERN - UAV & ROBOTICS

Computer Science student specializing in Artificial Intelligence with hands-on experience in machine learning, deep learning, computer vision, robotics servo control, camera-based automation, and AI model improvement. Passionate about UAV perception, autonomous robotics, and real-time intelligent automation.

EDUCATION

Saigon International University (SIU)

2023 - 2027

Bachelor's Degree in Computer Science - Artificial Intelligence Specialization

Relevant Coursework: Robotics, Machine Learning, Deep Learning, Computer Vision, MLOps, Data Structures, Python Programming, Database Systems, Software Engineering.

SELECTED PROJECTS

1. Autonomous Line-Following and Traffic Sign Recognition Robot on TurtleBot3 Burger

2025

TurtleBot3 Burger, ROS/ROS 2, Gazebo, RViz, Python, OpenCV, YOLOv8, Camera, Linux/Ubuntu

- Built an autonomous TurtleBot3 Burger system for lane following and traffic sign recognition in simulation and real-world testing.
- Developed real-time Python/OpenCV line detection and fine-tuned YOLOv8 for traffic sign detection.
- Integrated ROS/ROS 2 motion control for forward, left/right turn, and stop actions; achieved approximately **95% recognition accuracy under low-light conditions**.

GitHub: github.com/truongpk-lab/turtlebot3_autonomous | Demo: youtube.com/watch?v=SHMDS82wgqU

2. 6-Servo Robotic Arm Control System with Arduino and Python GUI

2025

Arduino Uno, Python, Tkinter, Serial Communication, Servo Motors, C/C++, pySerial

- Designed a 6-servo robotic arm using Arduino Uno and a Python/Tkinter desktop GUI.
- Programmed serial-based firmware and GUI controls for servo selection, angle control, gripper actions, and state display.
- Added two-way communication, servo limits, calibration, and manual action recording for movement reuse.

3. IoT-Based Personal Health Monitoring and Alert System Using Wearable Devices

Apr 2026

IoT, Wearable Sensors, Python, ML, MLOps, Backend API, Database, Health Dashboard

- Built an IoT health monitoring system using wearable sensor data such as acceleration, heart rate, sleep, and activity status.
- Trained models for fall detection, health risk assessment, and sleep quality analysis with time-series preprocessing.
- Combined acceleration and vital signs to reduce false fall detection; improved reliability by nearly **20%** over motion-only methods and achieved over **98%** in personalized risk evaluation during testing.

GitHub: [lot_simulator/HealthGuard](https://github.com/lot_simulator/HealthGuard) | Demo: youtube.com/watch?v=aHglab6YYII

4. Hand Gesture-Based Computer Mouse Control System

May 2026

Python, OpenCV, MediaPipe, PyAutoGUI, Hand Tracking, Gesture Recognition, Tkinter/PyQt

- Built a real-time hand gesture system for webcam-based mouse control using OpenCV, MediaPipe, and PyAutoGUI.
- Mapped hand landmarks to cursor movement, click, drag-and-drop, scroll, and interface control actions.
- Optimized smoothing, response speed, and configurable modes for office, entertainment, and quick actions.

GitHub: github.com/truongpk-lab/DieuKienBangTay

EXPERIENCE

AI Engineer Intern - My Phuoc Industrial Equipment Co., Ltd.

Jun 2025 - Sep 2025

Face Recognition-Based Attendance System | Python, OpenCV, DeepFace, ML, SQLite/MySQL, Flask/FastAPI, Git

- Built an automatic face recognition attendance system, including employee dataset collection, image preprocessing, recognition pipeline support, and attendance storage.
- Tested in real-world conditions and improved stability; achieved approximately **85-90% accuracy** under stable lighting with **1-3 seconds** processing time per check-in.